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Non-linear interactions in imperfect beams at veering

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Abstract

Non-linear interactions in a hinged–hinged uniform moderately curved beam with a torsional spring at one end are investigated. The two-mode interaction is a one-to-one autoparametric resonance activated in the vicinity of veering of the frequencies of the lowest two modes and results from the non-linear stretching of the beam centerline. The excitation is a base acceleration that is involved in a primary resonance with either the first mode only or with both modes. The ensuing non-linear responses and their stability are studied by computing force– and frequency–response curves via bifurcation analysis tools. Both the sensitivity of the internal resonance detuning—the gap between the veering frequencies—and the linear modal structure are investigated by varying the rise of the beam half-sinusoidal rest configuration and the torsional spring constant. The internal and external resonance detunings are varied accordingly to construct the non-linear system response curves. The beam mixed-mode response is shown to undergo several bifurcations, including Hopf and homoclinic bifurcations, along with the phenomenon of frequency island generation and mode localization.
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1. Introduction

The phenomena of frequency veering and mode localization have attracted considerable attention from solid state physics [1] to structural dynamics [2–9]. In general, when two frequencies approach each other as a control parameter is varied, there are two possibilities: they either intersect transversely entailing *crossover* or veer away from each other with high local curvature entailing the so-called *loci* or *curve veering*.

Crossovers typically occur in symmetric structural systems when variation of a control parameter entails that one frequency changes with a much greater rate than some other frequency. For example, in fixed–fixed first-mode buckled beams or in shallow arches with fixed hinges, the frequencies of the symmetric modes increase rapidly with the end load or the rise,

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respectively, due to the stiffening stretching mechanism whose influence grows with the magnitude of the initial curvature. On the other hand, the skew-symmetric modes do not experience stretching but are purely bending modes and are not affected by variations of the mentioned control parameters. Hence, in such systems crossovers occur between the frequencies of symmetric and skew-symmetric modes [10].

Leissa was the first to assert [2] that curve veering between the frequencies is due to the discretization of spatially continuous systems and discussed, in particular, membranes and plates. Leissa [2] and others also observed that, when curve veering occurs, the eigenfunctions undergo significant changes in the region of veering.

On the other hand, Perkins and Mote [3] showed, using a perturbation approach, that curve veering occurs in continuous as well as in discrete systems. Besides other structural systems, it was shown that a suspended cable with supports at different levels possesses eigenfunctions which are a mixture of symmetric and skew-symmetric components. The curve veering phenomenon occurs between the frequencies of the lowest two modes upon variation of a modified cable parameter that accounts for its elasto-geometric properties, including the inclined geometry. Recently, using a finite-difference scheme, Narakorn et al. [4] investigated the modal interactions occurring at the first three veering points of inclined cables undergoing free undamped vibrations.

An interesting aspect closely related to veering is mode localization. In linear periodic structures [5–8], mode localization is known as the phenomenon of confinement of the vibrational response within a small region of the overall structure through small variations in the geometric/mechanical parameters regulating its periodicity. In general, it is expected that the degree of localization depends on the magnitude of structural irregularity or mistuning. In one-dimensional continuous systems, such as strings or beams, Hodges [5] showed that localization is related to the reflection of waves by the constraints. On the other hand, in discrete systems, such as an array of weakly coupled pendula, the phenomenon is due to the fact that, while one of the pendula can oscillate close to its natural frequency, the others—due to the disordered structure—are driven off resonance and respond with much smaller amplitudes.

Pierre and Dowell [6,7] studied the localization phenomenon in a number of linear structural elements using a perturbation approach and showed a connection between curve veering and mode localization. Later, Triantafyllou and Triantafyllou [8] developed a unified geometrical description of frequency coalescence and curve veering, and then related their common geometric structure to the issue of localization in conservative and dissipative systems undergoing free vibrations.

Chan and Liu [9] showed that practical engineering structures such as aircrafts, although commonly idealized as symmetric, because of manufacturing and material tolerances and assemblage technology, exhibit asymmetric left and right sides. Using five simplified analytical models—three T-type beam models of horizontal tail with rear fuselage, a model of two coupled control surfaces and a model of an aircraft with external stores—they documented mode localization and frequency loci veering.

Relatively recent papers [11,12] have shown that localization occurs in weakly coupled non-linear cyclic structures consisting of discrete oscillators. In particular, King and Vakakis [12] have demonstrated the existence of non-linear mode localization for coupled geometrically non-linear beams in the absence of structural disorder. These studies have generally focused on perfectly tuned systems with weak symmetric (odd) non-linear grounding stiffness. To bridge the gap between results from linear and non-linear localization, King and Layne [13] investigated the combined effects of structural irregularities and non-linearities on the unforced dynamics of weakly coupled N-degree-of-freedom cyclic systems with cubic grounding stiffness. They showed that these cyclic assemblies exhibit a combination of localized and non-localized non-linear normal modes exceeding the number of linear normal modes. Further, they ascertained that for largely mistuned systems, only strongly localized motions are physically realized.

Nonetheless, most of the reviewed research has primarily addressed linear unforced structural vibrations and, to a smaller extent, non-linear unforced vibrations of weakly coupled disordered structures. Then, the question naturally arises on how the presence of the inherent structural non-linearities—responsible for coupling the modes in a non-linear sense—can affect these phenomena in forced spatially continuous imperfect systems and, hence, how the response

scenarios evolve on entering the non-linear vibration regime.

It is known that internal resonances, also known as autoparametric resonances, occur in forced damped or in unforced undamped non-linear systems as a result of the non-linear coupling forces [14]. For example, internal resonances have been investigated in suspended cables at first crossover [15] where the first symmetric in-plane mode is involved in a one-to-one internal resonance with the first skew-symmetric in-plane and out-of-plane modes and, simultaneously, in a two-to-one internal resonance with the first symmetric out-of-plane mode. As a result of these non-linear interactions, rich dynamics are exhibited by the cable with highly complex responses.

Tien et al. [16,17] studied the local and global bifurcations of a planar hinged-hinged symmetric shallow arch with one-to-one and two-to-one internal resonances. Again, the one-to-one resonance was at the crossover point between the frequencies of the lowest two modes. They numerically found a period-doubling route to chaos involving the two-period quasiperiodic coupled-mode motions and obtained analytical conditions for the onset of a Smale horseshoe-type chaos through Silnikov phenomenon.

In this paper, we investigate the non-linear one-to-one interactions excited by an external primary-resonance base acceleration of a hinged-hinged imperfect beam with a torsional spring at one end and possessing veering between the frequencies of the lowest two modes. The paper is organized as follows. In the next section, the general formulation and the linear eigenvalue problem are presented. In Section 3, the perturbation analysis of one-to-one autoparametric resonances at veering is summarized. In Section 4, the main outcomes of the bifurcation analyses are reported. Finally, in Section 5, the concluding remarks are drawn.

2. Problem formulation

Moderately large transverse vibrations of a hinged–hinged imperfect elastic beam, with a linear torsional spring at one of its end sections, around the curved rest configuration $\hat{\psi}$, are governed, in dimensional form, by the following integral-partial-differential equation [18]:

$$\begin{aligned} m\frac{\partial^2\hat{w}}{\partial\hat{t}^2} + EI\frac{\partial^4\hat{w}}{\partial\hat{x}^4} - \frac{EA}{\ell}\frac{\mathrm{d}^2\hat{\psi}}{\mathrm{d}\hat{x}^2}\int_0^\ell \frac{\partial\hat{w}}{\partial\hat{x}}\frac{\mathrm{d}\hat{\psi}}{\mathrm{d}\hat{x}}\,\mathrm{d}\hat{x} \\ = \frac{EA}{\ell}\frac{\partial^2\hat{w}}{\partial\hat{x}^2}\int_0^\ell \frac{\partial\hat{w}}{\partial\hat{x}}\frac{\mathrm{d}\hat{\psi}}{\mathrm{d}\hat{x}}\,\mathrm{d}\hat{x} \\ + \frac{EA}{2\ell}\frac{\mathrm{d}^2\hat{\psi}}{\mathrm{d}\hat{x}^2}\int_0^\ell \left(\frac{\partial\hat{w}}{\partial\hat{x}}\right)^2\mathrm{d}\hat{x} \\ + \frac{EA}{2\ell}\frac{\partial^2\hat{w}}{\partial\hat{x}^2}\int_0^\ell \left(\frac{\partial\hat{w}}{\partial\hat{x}}\right)^2\mathrm{d}\hat{x} - \hat{c}\frac{\partial\hat{w}}{\partial\hat{t}} - m\frac{\partial^2\hat{z}}{\partial\hat{t}^2} \end{aligned} \tag{1}$$

supplemented with the boundary conditions

$$\hat{w} = 0, \quad EI\frac{\partial^2\hat{w}}{\partial\hat{x}^2} - K\frac{\partial\hat{w}}{\partial\hat{x}} = 0 \quad \text{at} \quad \hat{x} = 0, \tag{2}$$

$$\hat{w} = 0, \quad EI\frac{\partial^2\hat{w}}{\partial\hat{x}^2} = 0 \quad \text{at} \quad \hat{x} = \ell, \tag{3}$$

where $\hat{w}(\hat{x},\hat{t})$ is the transverse deflection measured from $\hat{\psi}(\hat{x})$; $\hat{t}$ is the dimensional time; $\hat{x}$ is the horizontal coordinate; m is the mass per unit length; ℓ is the span of the beam; A and I denote the area and principal moment of inertia of the cross section, respectively; $\hat{c}$ is the coefficient of linear viscous damping; E is Young's modulus; K is the constant of the torsional spring attached to the beam section at $\hat{x} = 0$; and $\partial\hat{z}^2/\partial\hat{t}^2$ is the uniform acceleration (external excitation) prescribed to the beam supports.

The employed mechanical model is based on (i) the shallowness of the beam curved rest configurations, (ii) linear bending curvature, (iii) static condensation of the longitudinal motion, and (iv) cubic geometric hardening non-linearity due to the beam centerline stretching on due consideration of the boundary conditions. The justification of assumption (iv) lies in the fact that previous works have demonstrated that the geometric non-linearity due to stretching dominates on other non-linearities such as non-linear curvature and inertia non-linearities. Due to assumption (iii), the longitudinal motion is then slave of the transverse motion and is expressed as

$$\begin{aligned} \hat{u}(\hat{x},\hat{t}) = \frac{\hat{x}}{\ell}\int_0^\ell \left[\frac{\mathrm{d}\hat{\psi}}{\mathrm{d}\hat{x}}\frac{\partial\hat{w}}{\partial\hat{x}} + \frac{1}{2}\left(\frac{\partial\hat{w}}{\partial\hat{x}}\right)^2\right]\mathrm{d}x \\ - \int_0^{\hat{x}} \left[\frac{\mathrm{d}\hat{\psi}}{\mathrm{d}\hat{x}}\frac{\partial\hat{w}}{\partial\hat{x}} + \frac{1}{2}\left(\frac{\partial\hat{w}}{\partial\hat{x}}\right)^2\right]\mathrm{d}x. \end{aligned} \tag{4}$$

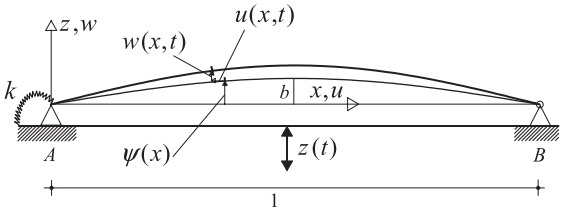

Fig. 1. The schematic imperfect curved beam geometry and the excitation.

With the adopted assumptions, Eqs. (1)–(3), governed by quadratic (due to the initial curvature) and cubic (due to the axis stretching) non-linearities, describe with acceptable accuracy the system behavior within the moderately large-amplitude vibration regime.

Next, we introduce the following non-dimensional variables and parameters (see Fig. 1 for the definitions of ψ, x, w, and $\ddot{z}$): $x=\hat{x}/\ell$, $\psi=\hat{\psi}/r$, $w=\hat{w}/r$, $z=\hat{z}/r$, and $t=\hat{t}\sqrt{EI/m\ell^4}$, where r is the radius of gyration of the cross section. Then, Eq. (1) is transformed into its simplest non-dimensional form as

$$\begin{aligned}\frac{\partial^2 w}{\partial t^2}+\frac{\partial^4 w}{\partial x^4}-\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}\int_0^1\frac{\partial w}{\partial x}\frac{\mathrm{d}\psi}{\mathrm{d}x}\,\mathrm{d}x\\ =\frac{\partial^2 w}{\partial x^2}\int_0^1\frac{\partial w}{\partial x}\frac{\mathrm{d}\psi}{\mathrm{d}x}\,\mathrm{d}x+\frac{1}{2}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}\int_0^1\left(\frac{\partial w}{\partial x}\right)^2\mathrm{d}x\\ +\frac{1}{2}\frac{\partial^2 w}{\partial x^2}\int_0^1\left(\frac{\partial w}{\partial x}\right)^2\mathrm{d}x-\varepsilon^2 c\frac{\partial w}{\partial t}\\ -\varepsilon^3 Z\cos\Omega t,\end{aligned}\tag{5}$$

where $\varepsilon^2 c=(\ell^2/\sqrt{mEI})\hat{c}$, $\varepsilon^3 Z$ is the amplitude of the base uniform harmonic acceleration, and ε denotes a small non-dimensional parameter used as an ordering device. The boundary conditions, in non-dimensional form, are

$$w=0\quad\text{and}\quad\frac{\partial^2 w}{\partial x^2}-k\frac{\partial w}{\partial x}=0\quad\text{at}\quad x=0,\tag{6}$$

$$w=0\quad\text{and}\quad\frac{\partial^2 w}{\partial x^2}=0\quad\text{at}\quad x=1,\tag{7}$$

where $k=K\ell/EI$ is the non-dimensional torsional spring constant and is assumed, in the subsequent perturbation analysis, to be O(1).

2.1. Natural frequencies and mode shapes

The linear unforced undamped vibration problem is obtained from Eq. (5) by dropping the damping term, the excitation, and the non-linear terms; that is,

$$\begin{aligned}\frac{\partial^2 w}{\partial t^2}+\mathscr{L}w=\frac{\partial^2 w}{\partial t^2}+\frac{\partial^4 w}{\partial x^4}\\ -\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}\int_0^1\frac{\partial w}{\partial x}\frac{\mathrm{d}\psi}{\mathrm{d}x}\,\mathrm{d}x=0\end{aligned}\tag{8}$$

subject to the boundary conditions (6) and (7). In Eq. (8), $\mathscr{L}$ denotes the linear positive-definite self-adjoint integral-differential stiffness operator. Consequently, the linear eigenvalue problem is self-adjoint; the eigenfunctions $\phi_m(x)$ are mutually orthogonal and are normalized so that $\int_0^1\phi_m\phi_n\,\mathrm{d}x=\langle\phi_m,\phi_n\rangle=\delta_{mn}$ and $\langle\phi_m,\mathscr{L}\phi_n\rangle=\omega_n^2\delta_{mn}$, where δ_{mn} is the Kronecker delta and use of the fact that the non-dimensional inertia operator is unitary has been made. For a beam with a rest curved configuration in the form of a half-sinusoidal wave, namely $\psi(x)=b\sin\pi x$, with $b=\hat{b}/r$, the eigenmodes can be expressed as

$$\begin{aligned}\phi_n(x)=c_1\cos\sqrt{\omega_n}x+c_2\sin\sqrt{\omega_n}x\\ +c_3\cosh\sqrt{\omega_n}x+c_4\sinh\sqrt{\omega_n}x\\ +c_5\sin\pi x\end{aligned}\tag{9}$$

and the associated natural frequencies ω_n are solutions of the algebraic eigenvalue problem reported in the Appendix.

In Fig. 2, we show variation of the non-dimensional natural frequencies with the rise level b when $k=1$. Veering between the natural frequencies of the first and second modes occurs when $b\approx 5.6$ and those of the second and third modes when $b\approx 12.7$. Further, two-to-one internal resonances may occur between the lowest two modes when $b\approx 2.4$ and $b\approx 11.4$ and between the second and third modes when $b\approx 6.2$. Also, three-to-one internal resonances may be activated between the lowest two modes when $b\approx 1.2$ and between the first and third modes when $b\approx 4.0$ and $b\approx 17.2$, respectively.

To gain insight into the linear modal structure around veering, we preliminarily note that the distributed elastic stiffness operator $\mathscr{L}$ comprises two terms; namely, $\mathscr{L}_f=\partial^4 w/\partial x^4$ is the bending stiffness term and $\mathscr{L}_s=b^2\pi^3\sin\pi x\int_0^1(\cos\pi x)(\partial w/\partial x)\,\mathrm{d}x$ is the stiffness term associated with the stretching

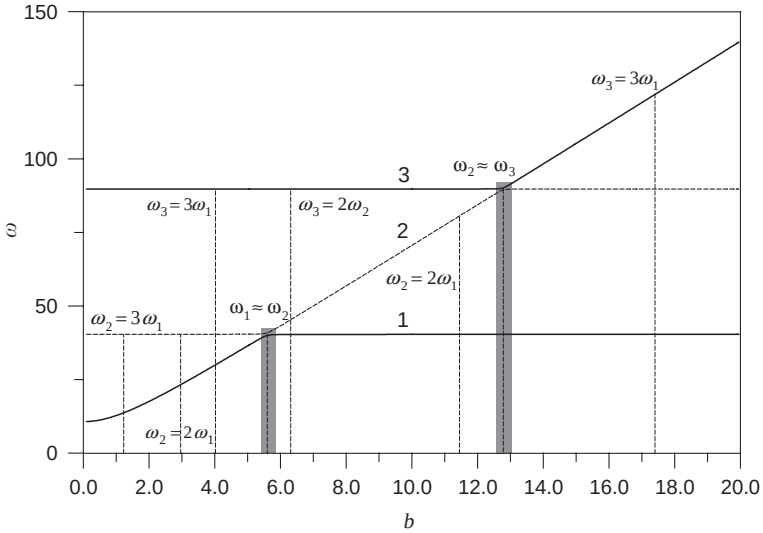

Fig. 2. Variation of the non-dimensional frequencies with the non-dimensional rise b when $k = 1$.

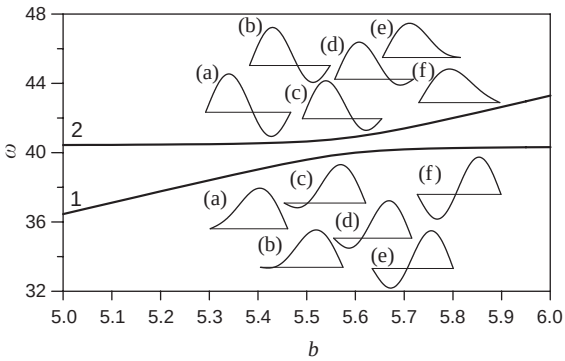

Fig. 3. Variation of the lowest two non-dimensional frequencies along with the associated mode shapes (bottom: first mode shape, top: second mode shape) with the non-dimensional rise b when $k = 1$. In particular, (a) $b = 5.40$, (b) $b = 5.50$, (c) $b = 5.55$, (d) $b = 5.60$, (e) $b = 5.70$, and (f) $b = 5.80$.

deformation mechanism. Hence, the latter depends on the square of the rise b and its influence is greater for predominantly symmetric modes due to the integral term multiplied by $\sin \pi x$. Solving the eigenvalue problem for various rise levels around the veering region yields the mode shapes of the veering modes shown in Fig. 3. When $b < 5.6$, the first mode, although dominated by the symmetric component, due to the action of the torsional spring at the left end section, is affected by a certain degree of asymmetry. On the other hand, the second mode is mostly skew-symmetric. This entails that the first mode is a stretching–bending mode and the second mode is an essentially bending mode. Around the veering region, $b \approx 5.6$, the skew-symmetric component of the first mode shape builds up as is clear from the growing region of opposite curvature on the left and concurrently the stretching-induced stiffness tends to increase with a smaller rate with increasing b. When $b > 5.6$, the skew-symmetric component of the first mode overpowers the symmetric component and the mode becomes a fundamentally bending mode whose frequency remains stationary with further increases of the rise level. A reverse process occurs for the second mode that gradually turns from a bending into a stretching–bending mode with increasing frequency due to the increasing stretching-induced stiffness.

Investigating the effects of the torsional spring constant variations around 1, it turned out that, with softer springs, $k < 1$, the gap between the veering frequencies decreases, as expected. In particular, the overall variation of the frequencies with the rise level was investigated in Fig. 4a considering a softer torsional spring, $k = 0.25$. The gaps between the frequencies of the first and second mode and those of the second and third mode become rather small entailing a

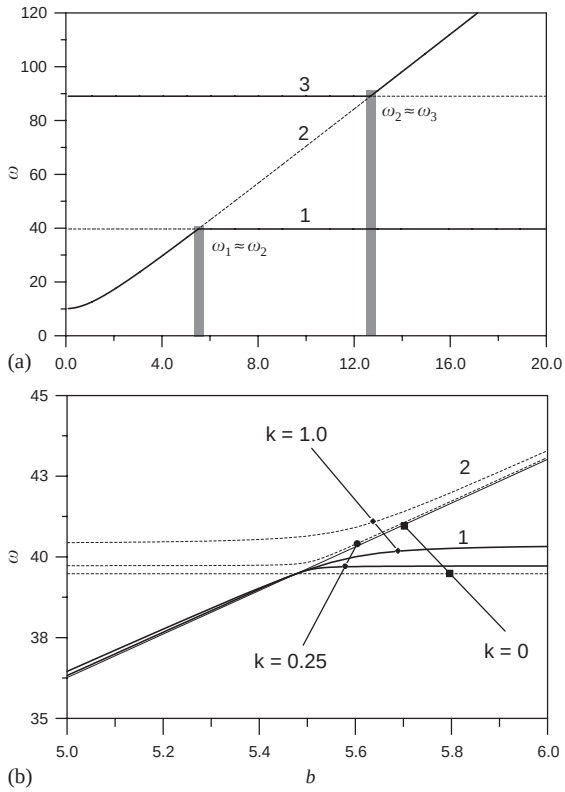

Fig. 4. Variation of the non-dimensional frequencies with the non-dimensional rise b when (a) $k=0.25$ and (b) $k=0.0$, $k=0.25$, and $k=1.0$.

near tangential intersection between the veering frequencies. The difference in gap between the frequencies of the lowest two modes when $k=1.0$ and 0.25, respectively, can be appreciated more closely in Fig. 4b. Clearly, in the limit case of vanishing torsional spring, $k=0$, the full symmetry is recovered, hence, the veering phenomenon is destroyed and crossovers occur between the frequencies with transversal intersection. As mentioned, the interactions occurring at the first crossover of a symmetric shallow arch were investigated in [10,16,17].

3. One-to-one internal resonances at veerings: perturbation analysis

General theoretical results relating to one-to-one interactions (i.e., the modulation equations and the expansions of the displacement and stress fields) activable in non-linear spatially continuous systems with arbitrary linear, quadratic, and cubic geometric operators were obtained in [19] by applying directly the method of multiple scales to the governing equations of motion and boundary conditions [19,20]. To use these general results in the present case, in agreement with the notation in [19,20], we recast the equation of motion and boundary conditions, Eqs. (5)–(7), in the following general operatorial format:

$$\ddot{w} + \mathscr{L} w = \mathscr{G}_2(w, w) + \mathscr{G}_3(w, w, w) - \varepsilon^2 c\, \dot{w} - \varepsilon^3 Z \cos \Omega t, \tag{10}$$

$$\mathscr{B}_{1i}(w) = 0 \quad \text{at} \quad x = 0 \quad \text{for } i = 1, 2, \tag{11}$$

$$\mathscr{B}_{2i}(w) = 0 \quad \text{at} \quad x = 1 \quad \text{for } i = 1, 2. \tag{12}$$

In Eq. (10), $\mathscr{L}$ is the linear self-adjoint operator defined in Eq. (8) and $\mathscr{G}_2$ and $\mathscr{G}_3$ are the quadratic and cubic geometric functional operators, respectively. Namely, $\mathscr{G}_2(u, v) = u''\langle v', \psi'\rangle + (1/2)\psi''\langle u', v'\rangle$ and $\mathscr{G}_3(u, v, w) = (1/2)u''\langle v', w'\rangle$. The non-linear operators are non-commutative; i.e., $\mathscr{G}_2(u, v) \neq \mathscr{G}_2(v, u)$.

Since we are interested in exploring one-to-one autoparametric responses excited by an external primary resonance and the ensuing resonant forces appear at third order, we have demoted the damping and external excitation to third order whereby an overall balance of the internal and external forces can be achieved.

The external excitation is assumed to inject energy into the system via a primary-resonance mechanism. The closeness of this resonance is expressed as $\Omega = \omega_m + \varepsilon^2\sigma_1$, where Ω is the non-dimensional excitation frequency. For one-to-one internal resonances taking place at veering points, the closeness of the veering frequencies is expressed as $\omega_n = \omega_m + \varepsilon^2\sigma_2$, with $\sigma_2 > 0$.

The asymptotic expansion of the transverse deflection of the beam can be expressed, to second order, as [19]

$$\begin{aligned} w(x,t) \approx\ & a_m \cos(\omega_m t + \beta_m)\phi_m(x) \\ & + a_n \cos(\omega_n t + \beta_n)\phi_n(x) \\ & + \frac{1}{2}a_m^2[\cos(2\omega_m t + 2\beta_m)\Psi_{mm}(x) \\ & + \chi_{mm}(x)] \\ & + \frac{1}{2}a_n^2[\cos(2\omega_n t + 2\beta_n)\Psi_{nn}(x) + \chi_{nn}(x)] \end{aligned}$$

$$+\frac{1}{2}a_n a_m\{\cos[(\omega_n+\omega_m)t+\beta_n+\beta_m]\Psi_{mn}(x)+\cos[(\omega_n-\omega_m)t+\beta_n-\beta_m]\chi_{mn}(x)\} \tag{13}$$

whereas the longitudinal motion, as mentioned, can be obtained by substituting Eq. (13) into the non-dimensional form of Eq. (4).

The second-order shape functions Ψ_{mm}, Ψ_{nn}, Ψ_{mn}, χ_{mm}, χ_{nn}, and χ_{mn} are solutions of the following boundary-value problems:

$$\mathscr{L}\Psi_{mm}-4\omega_m^2\Psi_{mm}=\mathscr{G}_2(\phi_m,\phi_m), \tag{14}$$

$$\mathscr{L}\Psi_{nn}-4\omega_n^2\Psi_{nn}=\mathscr{G}_2(\phi_n,\phi_n), \tag{15}$$

$$\mathscr{L}\Psi_{mn}-(\omega_m+\omega_n)^2\Psi_{mn}=\mathscr{G}_2(\phi_m,\phi_n)+\mathscr{G}_2(\phi_n,\phi_m), \tag{16}$$

$$\mathscr{L}\chi_{mm}=\mathscr{G}_2(\phi_m,\phi_m), \tag{17}$$

$$\mathscr{L}\chi_{nn}=\mathscr{G}_2(\phi_n,\phi_n), \tag{18}$$

$$\mathscr{L}\chi_{mn}=\mathscr{G}_2(\phi_m,\phi_n)+\mathscr{G}_2(\phi_n,\phi_m), \tag{19}$$

where the Ψ_{ij} and χ_{ij} satisfy the boundary conditions given by Eqs. (6) and (7). The boundary-value problems were solved in closed form when $b=5.6$ and $k=1$ using the pertinent mode shapes, ϕ_m and ϕ_n, shown in Fig. 3d. The obtained six shape functions are shown in Fig. 5; clearly, as expected they are neither symmetric nor skew-symmetric.

In Eq. (13), the amplitudes a_j and phases β_j are solutions of the following modulation equations that govern the slow flow of the beam dynamics:

$$\begin{aligned}2\mathrm{i}\omega_m\left(\frac{\mathrm{d}A_m}{\mathrm{d}t}+\mu_m A_m\right)&=K_{mm}A_m^2\bar{A}_m+K_3A_n^2\bar{A}_m\mathrm{e}^{2\mathrm{i}\sigma_2 t}+K_{mn}A_mA_n\bar{A}_n\\&\quad+2K_2A_m\bar{A}_mA_n\mathrm{e}^{\mathrm{i}\sigma_2 t}+K_2A_m^2\bar{A}_n\mathrm{e}^{-\mathrm{i}\sigma_2 t}\\&\quad+K_1A_n^2\bar{A}_n\mathrm{e}^{\mathrm{i}\sigma_2 t}-\frac{1}{2}P_m\mathrm{e}^{\mathrm{i}\sigma_1 t},\end{aligned} \tag{20}$$

$$\begin{aligned}2\mathrm{i}\omega_n\left(\frac{\mathrm{d}A_n}{\mathrm{d}t}+\mu_n A_n\right)&=K_2A_m^2\bar{A}_m\mathrm{e}^{-\mathrm{i}\sigma_2 t}\\&\quad+K_1A_n^2\bar{A}_m\mathrm{e}^{\mathrm{i}\sigma_2 t}+2K_1A_mA_n\bar{A}_n\mathrm{e}^{-\mathrm{i}\sigma_2 t}\\&\quad+K_{mn}A_m\bar{A}_mA_n+K_3A_m^2\bar{A}_n\mathrm{e}^{-2\mathrm{i}\sigma_2 t}\\&\quad+K_{nn}A_n^2\bar{A}_n-\frac{1}{2}P_n\mathrm{e}^{\mathrm{i}(\sigma_1-\sigma_2)t},\end{aligned} \tag{21}$$

where

$$\begin{aligned}P_k&=\int_0^1 Z\phi_k(x)\,\mathrm{d}x=\langle\phi_k,Z\rangle,\\2\mu_k&=\int_0^1 c\phi_k^2(x)\,\mathrm{d}x,\quad\text{for}\quad k=m,n.\end{aligned} \tag{22}$$

Moreover,

$$\begin{aligned}A_m(t)&=\frac{1}{2}a_m(t)\mathrm{e}^{\mathrm{i}[\beta_m(t)+\sigma_1 t]},\\A_n(t)&=\frac{1}{2}a_n(t)\mathrm{e}^{\mathrm{i}[\beta_n(t)+(\sigma_1-\sigma_2)t]}\end{aligned} \tag{23}$$

are the complex-valued amplitudes of the modes at their frequencies, respectively, and $\bar{A}_m(t)$ and $\bar{A}_n(t)$ are their complex conjugates. The non-linear interaction coefficients are expressed as [19]:

$$\begin{aligned}K_{hh}&=\langle\phi_h,\mathscr{G}_2(\phi_h,\Psi_{hh})\rangle+\langle\phi_h,\mathscr{G}_2(\Psi_{hh},\phi_h)\rangle\\&\quad+2\langle\phi_h,\mathscr{G}_2(\phi_h,\chi_{hh})\rangle+2\langle\phi_h,\mathscr{G}_2(\chi_{hh},\phi_h)\rangle\\&\quad+3\langle\phi_h,\mathscr{G}_3(\phi_h,\phi_h,\phi_h)\rangle,\quad\text{for } h=m,n,\end{aligned} \tag{24}$$

$$\begin{aligned}K_{mn}&=\langle\phi_m,\mathscr{G}_2(\phi_n,\Psi_{mn})\rangle+\langle\phi_m,\mathscr{G}_2(\Psi_{mn},\phi_n)\rangle\\&\quad+\langle\phi_m,\mathscr{G}_2(\phi_n,\chi_{mn})\rangle+\langle\phi_m,\mathscr{G}_2(\chi_{mn},\phi_n)\rangle\\&\quad+2\langle\phi_m,\mathscr{G}_2(\phi_m,\chi_{nn})\rangle+2\langle\phi_m,\mathscr{G}_2(\chi_{nn},\phi_m)\rangle\\&\quad+2\langle\phi_m,\mathscr{G}_3(\phi_n,\phi_n,\phi_m)\rangle\\&\quad+2\langle\phi_m,\mathscr{G}_3(\phi_n,\phi_m,\phi_n)\rangle\\&\quad+2\langle\phi_m,\mathscr{G}_3(\phi_m,\phi_n,\phi_n)\rangle,\end{aligned} \tag{25}$$

$$\begin{aligned}K_1&=\langle\phi_n,\mathscr{G}_2(\phi_m,\Psi_{nn})\rangle+\langle\phi_n,\mathscr{G}_2(\Psi_{nn},\phi_m)\rangle\\&\quad+\langle\phi_n,\mathscr{G}_2(\phi_n,\chi_{mn})\rangle+\langle\phi_n,\mathscr{G}_2(\chi_{mn},\phi_n)\rangle\\&\quad+\langle\phi_n,\mathscr{G}_3(\phi_n,\phi_n,\phi_m)\rangle+\langle\phi_n,\mathscr{G}_3(\phi_n,\phi_m,\phi_n)\rangle\\&\quad+\langle\phi_n,\mathscr{G}_3(\phi_m,\phi_n,\phi_n)\rangle,\end{aligned} \tag{26}$$

$$\begin{aligned}K_2&=\langle\phi_n,\mathscr{G}_2(\phi_m,\Psi_{mm})\rangle+\langle\phi_n,\mathscr{G}_2(\Psi_{mm},\phi_m)\rangle\\&\quad+2\langle\phi_n,\mathscr{G}_2(\phi_m,\chi_{mm})\rangle+2\langle\phi_n,\mathscr{G}_2(\chi_{mm},\phi_m)\rangle\\&\quad+3\langle\phi_n,\mathscr{G}_3(\phi_m,\phi_m,\phi_m)\rangle,\end{aligned} \tag{27}$$

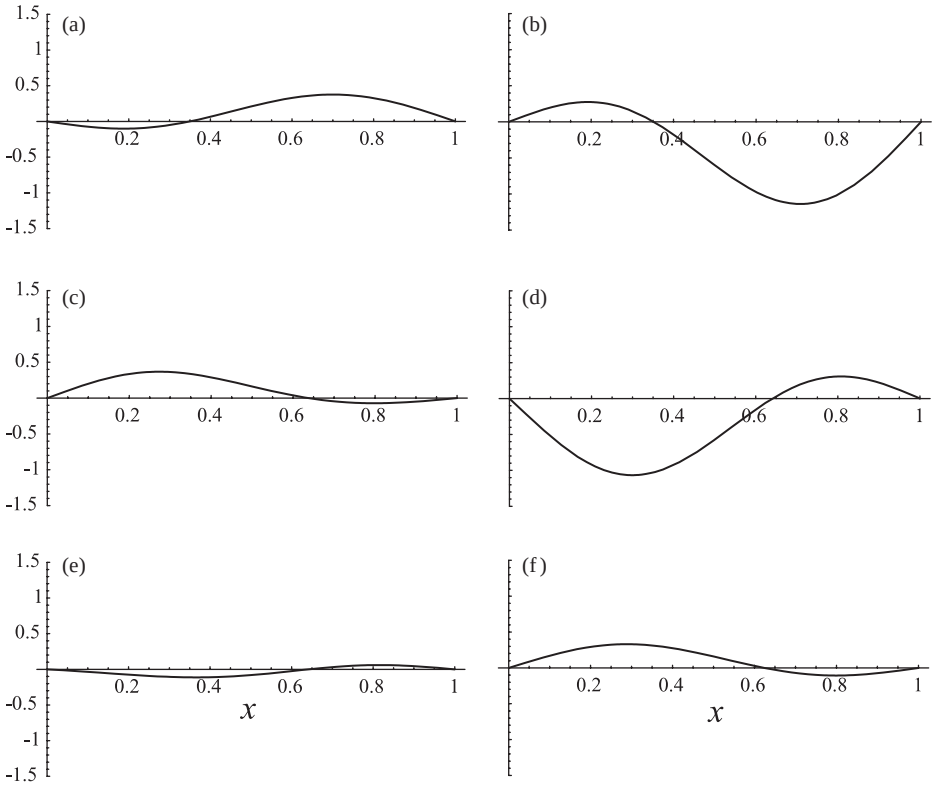

Fig. 5. Second-order shape functions when $b = 5.6$ and $k = 1$: (a) Ψ_{mm}, (b) χ_{mm}, (c) Ψ_{nn}, (d) χ_{nn}, (e) Ψ_{mn}, and (f) χ_{mn}.

$$\begin{aligned} K_3 = {} & \langle \phi_n, \mathscr{G}_2(\phi_n, \Psi_{mm})\rangle + \langle \phi_n, \mathscr{G}_2(\Psi_{mm}, \phi_n)\rangle \\ & + \langle \phi_n, \mathscr{G}_2(\phi_m, \chi_{mn})\rangle + \langle \phi_n, \mathscr{G}_2(\chi_{mn}, \phi_m)\rangle \\ & + \langle \phi_n, \mathscr{G}_3(\phi_m, \phi_m, \phi_n)\rangle + \langle \phi_n, \mathscr{G}_3(\phi_m, \phi_n, \phi_m)\rangle \\ & + \langle \phi_n, \mathscr{G}_3(\phi_n, \phi_m, \phi_m)\rangle. \end{aligned} \tag{28}$$

The time-averaged Lagrangian and virtual work term corresponding to Eqs. (20) and (21) are

$$\begin{aligned} \langle L\rangle = {} & \frac{1}{2}K_{mm}A_m^2\bar{A}_m^2 + \frac{1}{2}K_{nn}A_n^2\bar{A}_n^2 \\ & + \frac{1}{2}K_3[A_n^2\bar{A}_m^2 e^{2i\sigma_2 t} + A_m^2\bar{A}_n^2 e^{-2i\sigma_2 t}] \\ & + K_{mn}A_m\bar{A}_m A_n\bar{A}_n \\ & + K_2[A_m\bar{A}_m^2 A_n e^{i\sigma_2 t} + A_m^2\bar{A}_m\bar{A}_n e^{-i\sigma_2 t}] \\ & + K_1[A_n^2\bar{A}_n\bar{A}_m e^{i\sigma_2 t} + A_n\bar{A}_n^2 A_m e^{-i\sigma_2 t}] \\ & + i\omega_m\left(A_m\frac{d\bar{A}_m}{dt} - \bar{A}_m\frac{dA_m}{dt}\right) \\ & + i\omega_n\left(A_n\frac{d\bar{A}_n}{dt} - \bar{A}_n\frac{dA_n}{dt}\right), \end{aligned} \tag{29}$$

$$\begin{aligned} \langle \delta\mathscr{W}\rangle = {} & 2i\omega_m\mu_m(\bar{A}_m\delta A_m - A_m\delta\bar{A}_m) \\ & + 2i\omega_n\mu_n(\bar{A}_n\delta A_n - A_n\delta\bar{A}_n) \\ & - \frac{1}{2}P_m[e^{-i\sigma_1 t}\delta A_m + e^{i\sigma_1 t}\delta\bar{A}_m] \\ & - \frac{1}{2}P_n[e^{-i(\sigma_1-\sigma_2)t}\delta A_m \\ & + e^{i(\sigma_1-\sigma_2)t}\delta\bar{A}_m]. \end{aligned} \tag{30}$$

Next, Eqs. (20) and (21) are expressed in terms of real-valued variables. Substituting the polar transfor-

mations (23) into Eqs. (20) and (21) yields

$$\begin{aligned}\dot{a}_m = &-\mu_m a_m - \frac{K_1}{8\omega_m} a_m^3 \sin(\beta_m - \beta_n) \\ &- \frac{K_2}{8\omega_m} a_m^2 a_n \sin(\beta_m - \beta_n) + \frac{P_m}{2\omega_m} \sin \beta_m \\ &- \frac{K_3}{8\omega_m} a_m a_n^2 \sin[2(\beta_m - \beta_n)], \end{aligned} \tag{31}$$

$$\begin{aligned}a_m \dot{\beta}_m = &-\sigma_1 a_m - \frac{K_{mm}}{8\omega_m} a_m^3 - \frac{K_{mn}}{8\omega_m} a_m a_n^2 \\ &- \frac{K_1}{8\omega_m} a_m^3 \cos(\beta_m - \beta_n) + \frac{P_m}{2\omega_m} \cos \beta_m \\ &- \frac{3K_2}{8\omega_m} a_m^2 a_n \cos(\beta_m - \beta_n) \\ &- \frac{K_3}{8\omega_m} a_m a_n^2 \cos[2(\beta_m - \beta_n)], \end{aligned} \tag{32}$$

$$\begin{aligned}\dot{a}_n = &-\mu_n a_n + \frac{K_1}{8\omega_n} a_n^2 a_m \sin(\beta_m - \beta_n) \\ &+ \frac{K_2}{8\omega_n} a_m^3 \sin(\beta_m - \beta_n) + \frac{P_n}{2\omega_n} \sin \beta_n \\ &+ \frac{K_3}{8\omega_n} a_n a_m^2 \sin[2(\beta_m - \beta_n)], \end{aligned} \tag{33}$$

$$\begin{aligned}a_n \dot{\beta}_n = &-(\sigma_1 - \sigma_2) a_n - \frac{K_{nn}}{8\omega_n} a_n^3 - \frac{K_{mn}}{8\omega_n} a_n a_m^2 \\ &- \frac{3K_1}{8\omega_n} a_n^2 a_m \cos(\beta_m - \beta_n) + \frac{P_n}{2\omega_n} \cos \beta_n \\ &- \frac{K_2}{8\omega_n} a_m^3 \cos(\beta_m - \beta_n) \\ &- \frac{K_3}{8\omega_n} a_n a_m^2 \cos[2(\beta_m - \beta_n)]. \end{aligned} \tag{34}$$

To determine the equilibrium solutions or fixed points of Eqs. (31)–(34), we set $\dot{a}_m = \dot{a}_n = 0$ and $\dot{\beta}_m = \dot{\beta}_n = 0$. We note that, regardless of whether P_n is zero or non-zero, the modulation equations do not admit single-mode fixed points when $K_2 \neq 0$, as it occurs in the present illustrative example of beams with a half-sinusoidal configuration. This is due to the term $K_2 A_m^2 \bar{A}_m \exp(-\mathrm{i}\sigma_2 t)$ in Eq. (21). Similarly, if the nth mode is directly excited, the mth mode must also be activated due to the presence of the term $K_1 A_n^2 \bar{A}_n \exp(\mathrm{i}\sigma_2 t)$ in Eq. (20). Therefore, the following general proposition can be made: *for imperfect beams with* $(K_1, K_2) \neq 0$, *non-trivial fixed points only, corresponding to two-mode periodic motions of the beam and their bifurcated solutions, can exist in the vicinity of veering*.

4. Bifurcation analysis

The fixed points of Eqs. (31)–(34) are determined numerically using a pseudo-arclength technique [21] and their stability is ascertained by evaluating the four eigenvalues of the 4×4 Jacobian matrix. The dynamic solutions are determined by solving a two-point boundary-value problem using a shooting method and numerical integration. The stability of the limit cycles is determined using Floquet theory.

Next, we discuss the main outcomes of the bifurcation analysis of mixed-mode non-linear responses that occur near the veering point involving the first ($m = 1$) and second ($n = 2$) modes. In particular, the bifurcations responsible for the appearance of quasiperiodic motions and the mode localization phenomenon are outlined. The values of the considered parameters and computed coefficients are $k = 1$, $b = 5.6$, $\omega_m = 40.00$, $\omega_n = 40.91$, $K_{mm} = 1104.50$, $K_{nn} = 1282.02$, $K_{mn} = -1308.18$, $K_1 = 177.85$, $K_2 = 651.94$, and $K_3 = -406.42$. The interaction coefficients have been calculated evaluating the integrals in Eqs. (24)–(28) accounting for the solutions of the integral-differential boundary-value problems in Eqs. (14)–(19) and the mode shapes in Fig. 3d.

As we already pointed out, the perturbation analysis is based on the assumption $k = \mathrm{O}(1)$ so as to have the torsional spring exerting its reactive elastic moment at first order. Hence, k was set to 1 and $b = 5.6$ followed consequently at the first veering. The detuning of the veering condition—gap between the linear frequencies of the interacting modes—can be achieved by varying independently k and b. However, the overall detuning is compounded in the parameter σ_2. Mention must be made of the fact that the effects of this detuning in the non-linear interaction coefficients appearing in the modulation equations are of higher order and, hence, are neglected in the analysis.

The force–response curves are discussed first. Setting $\sigma_1 = 0$, $\mu_1 = 0.006$, and $\mu_2 = 0.008$ and putting $P_2 = 0$ and $P_1 = \langle Z, \phi_1 \rangle$, we illustrate the influence of the excitation amplitude Z on the response in the force-response curves in Fig. 6, obtained when $\sigma_2 = 0.05$. The solid lines indicate stable fixed points, the dashed

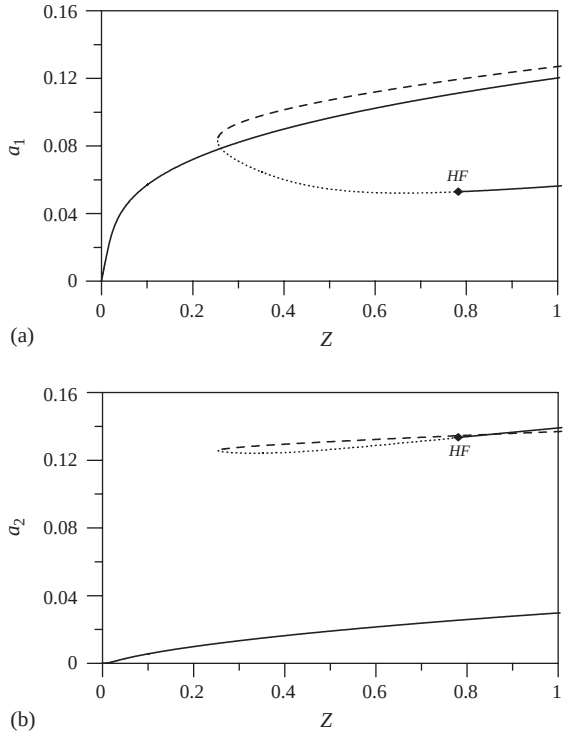

Fig. 6. Force–response curves when $\sigma_2 = 0.05$, $\sigma_1 = 0$, $P_2 = 0$, $\mu_1 = 0.006$, and $\mu_2 = 0.008$.

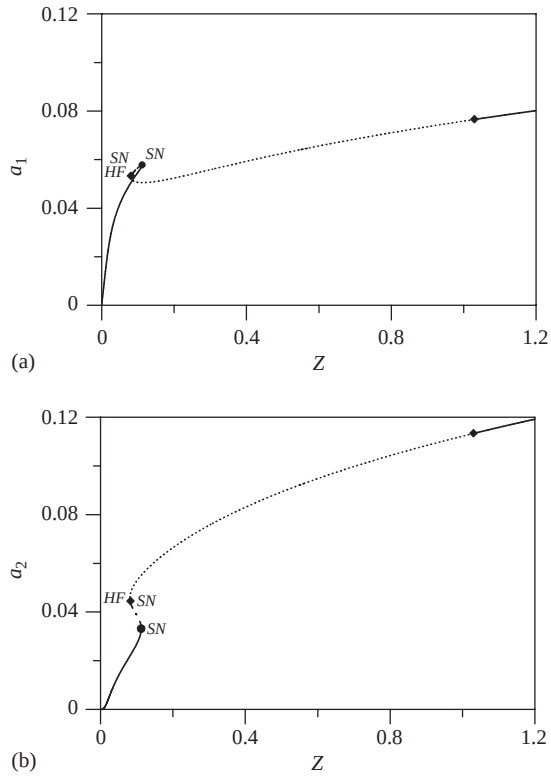

Fig. 7. Force–response curves when $\sigma_2 = 0$, $\sigma_1 = 0$, $P_2 = 0$, $\mu_1 = 0.006$, and $\mu_2 = 0.008$.

lines indicate saddles, and the dotted lines indicate unstable foci. As Z is increased from zero, stable fixed points appear and increase monotonously. For Z approximately greater than 0.25, a second branch exists. As Z is decreased from 1, the stable fixed points on this branch lose stability via a Hopf bifurcation near $Z \approx 0.8$. Although, both branches appear to be isolated from each other, one is a continuation of the other because they connect through a saddle–node bifurcation at a large non-physical value of Z, namely, $Z \approx 23.56$.

The influence of the internal detuning on the force–response curves was also investigated. Considering the limiting value of $\sigma_2 = 0$, corresponding to the limit case of very weak torsional spring, and keeping all other parameters unchanged, we computed the force–response curves shown in Fig. 7. In this case, the stable branch of fixed points emerging from the origin undergoes a saddle–node bifurcation as Z is increased. To the left of this saddle–node point, immediately past a second saddle–node, a Hopf bifurcation occurs—the closeness of these bifurcations is such that the branch of interposed stable solutions cannot be appreciated in the plot—resulting in limit cycles that coexist with the stable fixed points. However, increasing Z beyond the saddle–node point, we found that stable fixed points do not exist and, therefore, periodic and aperiodic solutions of the modulation equations are the only possible attractors. These solutions (two-period quasiperiodic and, possibly, aperiodic motions) persist up to $Z \approx 1$, where a reverse Hopf bifurcation occurs giving rise to stable fixed points.

The progression of the force-responses curves with σ_2 varying between 0 and 0.01 is presented in Fig. 8. Here, it is clear that the saddle–node bifurcation SN is very sensitive to small changes in σ_2 and continues to occur at larger values of Z as σ_2 increases. On the other hand, the Hopf bifurcation HF does not vary significantly with σ_2 and continues to occur at smaller val-

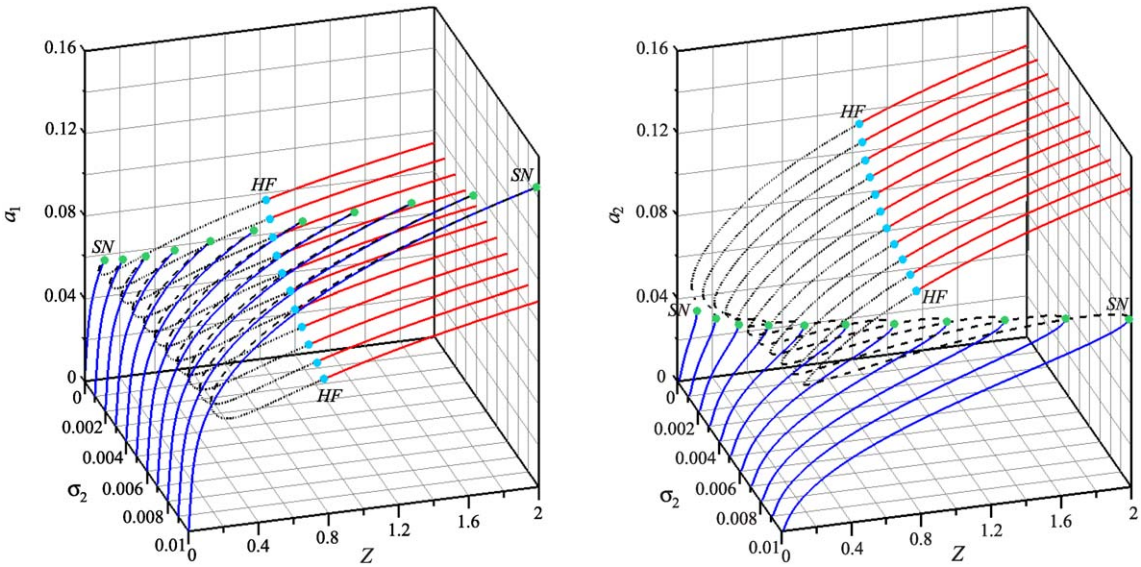

Fig. 8. Variation of the force–response curves with $\sigma_2 \in [0, 0.01]$ and $Z \in [0, 2]$ when $\sigma_1 = 0$, $P_2 = 0$, $\mu_1 = 0.006$, and $\mu_2 = 0.008$.

ues of Z as σ_2 increases. This behavior means that the region between SN and HF in Fig. 7, where no stable fixed points exist, becomes gradually smaller as σ_2 increases. Therefore, increasing σ_2 results in marginalizing any quasiperiodic or aperiodic solutions because of the continuous presence of a stable equilibrium solution, as in Fig. 6.

Next, we investigated the influence of the excitation frequency detuning, σ_1, on the fixed points as it is done experimentally when frequency sweeps are performed to construct the system frequency–response curves. We set $Z = 0.3$, $\mu_1 = 0.006$, $\mu_2 = 0.008$, and $\sigma_2 = 0.05$ and obtained the frequency–response curves shown in Fig. 9. There are two separate sets of curves. The main set, which spans the entire range of σ_1, is bent to the left, indicating a softening behavior. The second set of curves is found to be isolated and stable fixed points on it are found to undergo a Hopf bifurcation, leading to limit cycles. The documented frequency island generation was recently encountered also in a single-degree-of-freedom piecewise linear suspension system [22] under primary resonance where the island was generated only within certain ranges of the system parameters. Furthermore, an isolated frequency–response branch was also found in a Jeffcott rotor under a one-to-one internal resonance [23] excited by an external subharmonic resonance of order one-half.

A remarkably interesting behavior was found when decreasing the internal detuning σ_2 to 0; the previous two sets of curves coalesce into a single set of curves, as shown in Fig. 10. The frequency–response curves still preserve the same bifurcation points, namely saddle–node and Hopf bifurcations.

To draw a more thorough picture of the system bifurcation scenarios, we also investigated the effect of small variations of the internal resonance detuning parameter σ_2, as shown in Fig. 11. It was found that starting from a very low value of σ_2, the system exhibits a limit cycle, shown in Fig. 12a, that is stable as we increase σ_2. Moreover, as shown in Fig. 13, its period T increases with σ_2 suggesting a homoclinic bifurcation; the steep increase of the period of the limit cycle as it approaches the saddle (Figs. 11 and 13) is evident. Beyond this point, the limit cycle is destroyed, and subsequently only fixed points exist (a blue sky catastrophe). The overall evolution of the limit cycles below the homoclinic bifurcation is illustrated in Fig. 12, where a sketch of the periodic (a_1–a_2)-orbit is shown. We note that parts (a)–(c) in Fig. 12 coincide with the similarly marked points in Fig. 13, relating the period to the associated limit cycles.

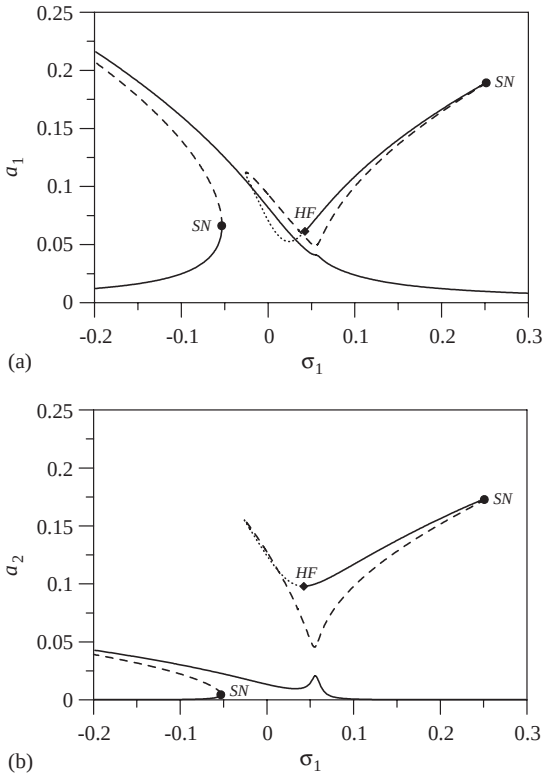

Fig. 9. Frequency–response curves when $\sigma_2 = 0.05$, $Z = 0.3$, $P_2 = 0$, $\mu_1 = 0.006$, and $\mu_2 = 0.008$.

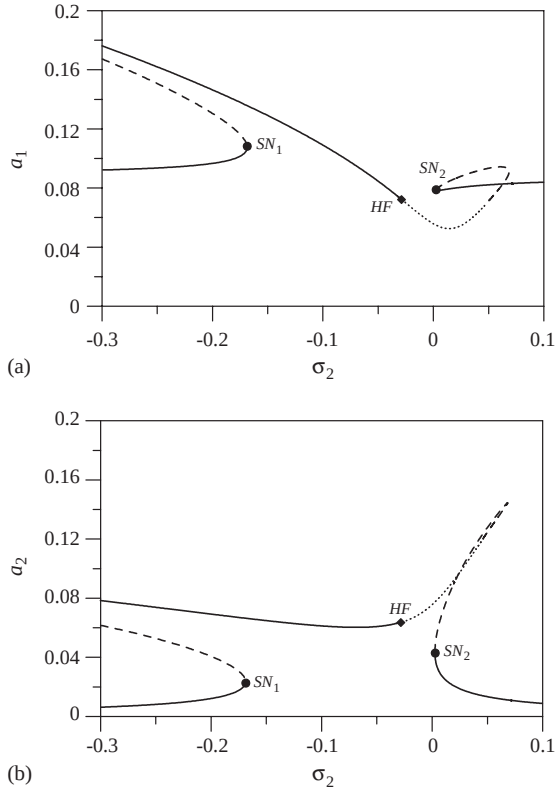

Fig. 10. Frequency–response curves when $\sigma_2 = 0$, $Z = 0.3$, $P_2 = 0$, $\mu_1 = 0.006$, and $\mu_2 = 0.008$.

It is here worth outlining the main differences and similarities between the bifurcation behavior of a perfect shallow curved beam at first crossover [16,17] and that of the investigated imperfect shallow curved beam at first veering. In the perfect case [17], the beam exhibits both uncoupled (low-frequency) and coupled-mode motions. On the other hand, the present imperfect beam possesses coupled-mode solutions only with possible localized motions as it will be further discussed in the next section. However, both the perfect and imperfect beams experience Hopf bifurcations causing the loss of stability of the coupled-mode solutions. Moreover, in the perfect case, Tien et al. [17] numerically demonstrated that the two-period quasiperiodic coupled-mode motions undergo a sequence of period-doubling bifurcations culminating into chaos.

4.1. Mode localization

As mentioned in Section 1, there is a close connection between mode localization and curve veering [6,7]. In the present investigation into the nonlinear resonant modal coupling occurring at veering, it was found that mode localization is manifested in a somewhat different fashion. The typical normal mode localization refers to the confinement of vibrational energy within rather small regions of the disordered systems. In the present case, it is found that the system may respond, depending on the initial conditions, with a dominant modal contribution from the first mode (see Figs. 3d and 6). Because this mode exhibits most of the vibrational energy distributed over three-fifths of the beam span, a great deal of the overall system response is localized in this region. The remaining region of the beam is affected by a smaller motion.

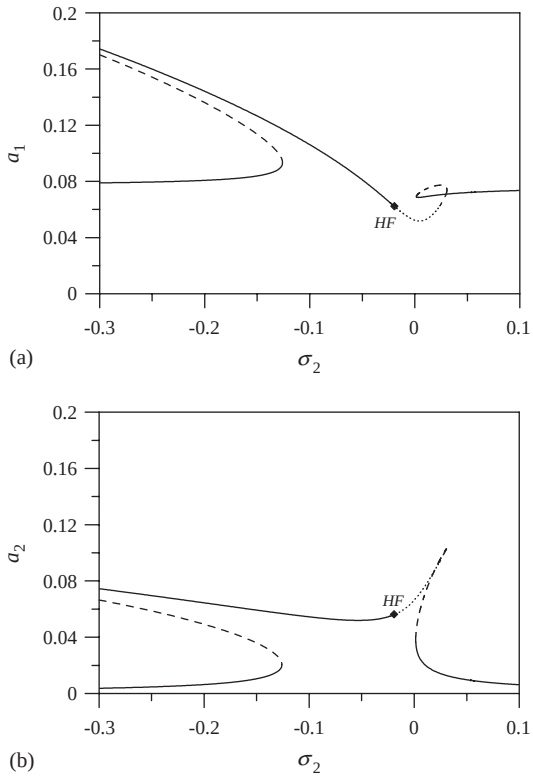

Fig. 11. Frequency–response curves when $P_2 = 0$, $Z = 0.3$, $\sigma_1 = 0$, $\mu_1 = 0.006$, and $\mu_2 = 0.008$. SN_1 and SN_2 denote saddle–node bifurcations and HF denotes a Hopf bifurcation.

Simultaneously, the beam may respond with comparable contributions from both modes (Figs. 3d and 6), which are complementary, thereby leading to a spreading of the vibrational energy onto the entire beam span.

In the force–response curves of Fig. 6 (when $\sigma_2 = 0.05$), it is clear that the amplitude of the directly excited mode ($P_1 \neq 0$ and $P_2 = 0$) is significantly higher than that of the indirectly excited second mode. This phenomenon seems to be destroyed by decreasing the internal resonance detuning σ_2, that is, decreasing the degree of mistuning (Fig. 7). However, considering an excitation possessing projections onto both modes (i.e., $P_1 \neq 0$ and $P_2 \neq 0$), the ensuing force–response curves in Figs. 14 and 15 show that *mode localization is robust with respect to changes in the internal detuning parameter*. In particular, in Fig. 13, we show the force-response curves when $\sigma_2 = 0.05$, $\sigma_1 = 0$, $\mu_1 = 0.006$, and $\mu_2 = 0.008$. As Z is increased from zero, both amplitudes increase monotonously, with a_1 being much larger that a_2. As Z is increased past 1, the response undergoes a Hopf bifurcation. A second branch of solutions where the amplitude a_2 is close to a_1 exists for approximately $Z > 0.2$.

Setting $\sigma_2 = 0$ and keeping all of the other parameters fixed, we show the corresponding force–response curves in Fig. 15. Here, it seems that the two branches of the previous figure coalesce and then switch such that the high-amplitude a_2-solution now originates from $Z = 0$ and the low-amplitude a_2-solution is now born out of a saddle–node bifurcation.

In Fig. 16, we show the beam response $w(x, t)$ over one period of oscillations, corresponding to the two coexisting stable solutions (A and B) in Fig. 14. In the left column (A), the response exhibits mode

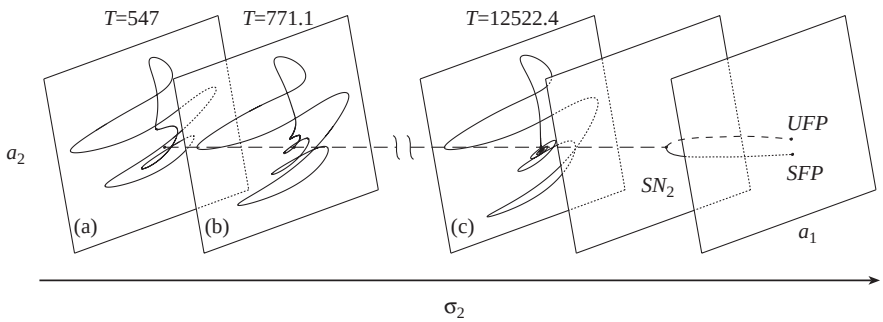

Fig. 12. Projections of the phase portraits onto the a_1–a_2 plane of dynamic solutions illustrating the homoclinic bifurcation when $P_2 = 0$, $Z = 0.3$, $\sigma_1 = 0$, $\mu_1 = 0.006$, and $\mu_2 = 0.008$. Parts (a)–(c) correspond to the similarly marked points in Fig. 13, SN_2 denotes a saddle–node bifurcation, and SFP and UFP denote stable and unstable fixed points, respectively.

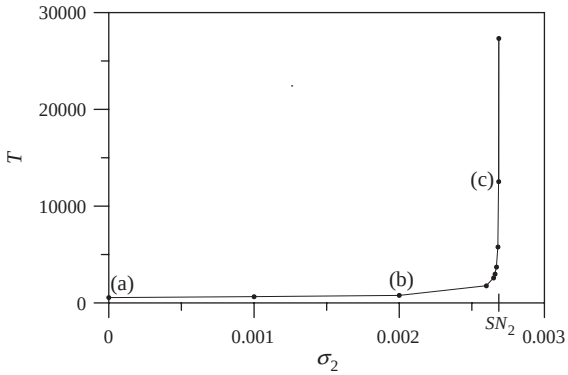

Fig. 13. Variation of the period T of the limit cycle as a function of σ_2 when $P_2 = 0$, $Z = 0.3$, $\sigma_1 = 0$, $\mu_1 = 0.006$, and $\mu_2 = 0.008$.

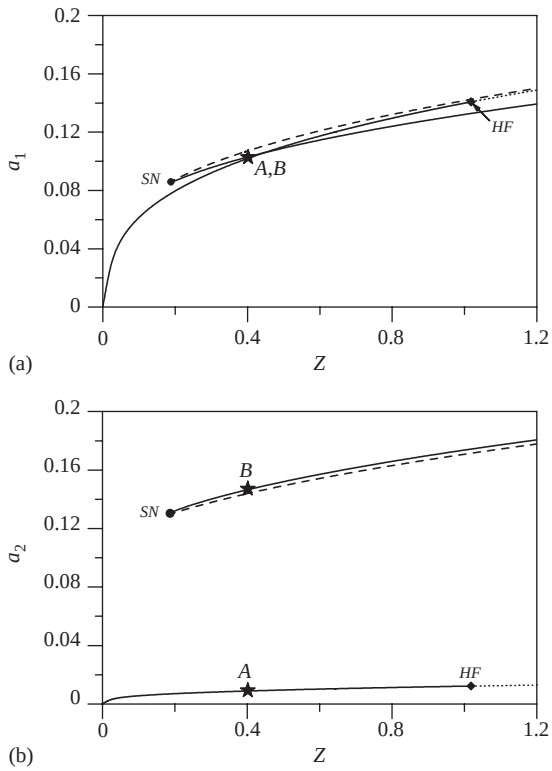

Fig. 14. Force–response curves when $\sigma_2 = 0.05$, $\sigma_1 = 0$, $\mu_1 = 0.006$, $\mu_2 = 0.008$, and with $(P_1,\ P_2) \neq 0$.

localization whereby most of the energy is confined within the first mode resulting in a "weakly localized" response [13]. In the right column (B), the coexisting response is a typical mixed-mode solution with com-

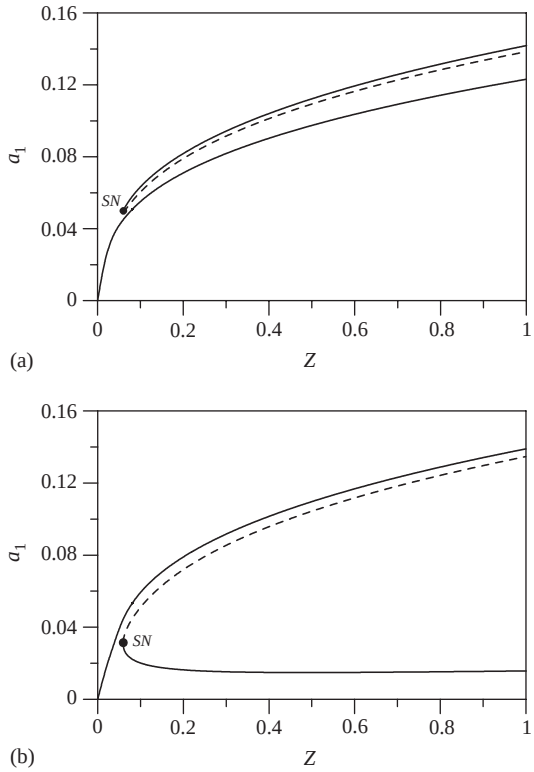

Fig. 15. Force–response curves when $\sigma_2 = 0$, $\sigma_1 = 0$, $\mu_1 = 0.006$, $\mu_2 = 0.008$, and with $(P_1,\ P_2) \neq 0$.

parable contributions from both modes resulting in a "spatially extended" motion [13].

Here, it is worth mentioning the investigations of King and Layne [13] on non-linear irregular cyclic systems, although relating, differently from the present case, to the unforced case and to symmetric non-linear grounding stiffness and weak linear coupling. For sufficiently strong non-linearity and/or sufficiently weak coupling stiffness, they obtained combinations of (i) extended, (ii) weakly localized, and (iii) strongly localized modes. For 3-dof systems, these modes can lose their stability via Hopf bifurcations. They also showed that, in the perfect case (absence of structural irregularity), the system possesses uncoupled modes that undergo pitchfork bifurcations leading to coupled modes. On the other hand, in the presence of mistuning, saddle–node bifurcations are responsible for the onset of coupled modes of types (i)–(iii).

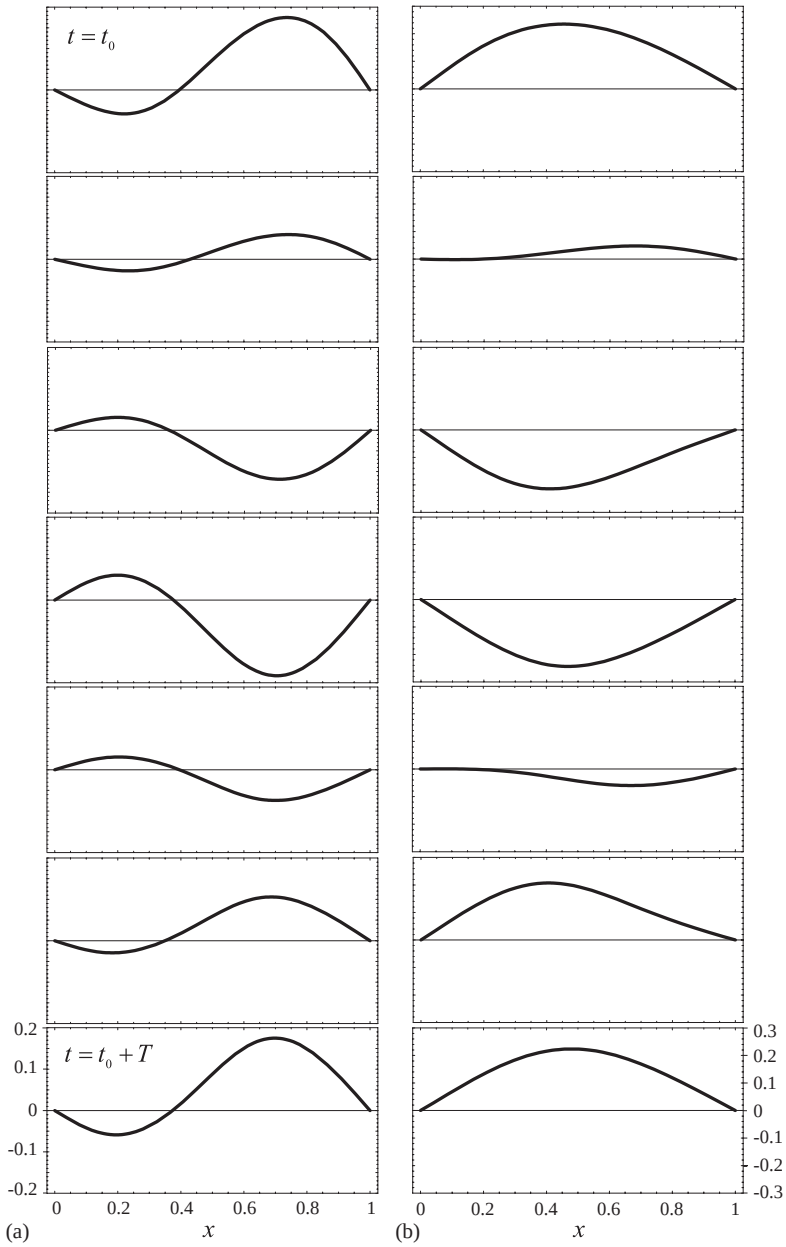

Fig. 16. Beam deflections $w(x,t)$, over one period of oscillations, of the coexisting stable responses marked A and B in Fig. 13 when $Z = 0.4$, $\sigma_1 = 0$, $\sigma_2 = 0.05$, $\mu_1 = 0.006$, $\mu_2 = 0.008$, and with $(P_1,\ P_2) \neq 0$.

5. Conclusions

The non-linear responses of a hinged–hinged uniform imperfect elastic beam with a torsional spring at one end have been investigated. For a torsional spring whose non-dimensional stiffness is of the same order of magnitude of the beam bending stiffness, the beam rise is selected such that veering between the frequencies of the lowest two modes occurs and, as a result of the non-linear stretching of the beam centerline, the

modes interact via a one-to-one autoparametric resonance mechanism. The energy is supplied by an external primary resonance of either the first mode only or of both modes.

A close analysis of the structure of the modulation equations governing the non-linear modal coupling (accounting for the calculated non-linear interaction coefficients) allows us to conclude that, in general, *only two-mode periodic motions of the beam and their bifurcated solutions can exist in the vicinity of veering* contrary to perfect shallow curved beams [16,17] where both uncoupled and coupled motions are physically realized.

The non-linear responses have been here studied using bifurcation analysis tools, leading to the construction of force– and frequency–response curves. The internal and external resonance detunings have been varied in the pertinent ranges to construct the response curves. *Hopf bifurcations* are found to be responsible for *two-period quasiperiodic responses* of the beam. The post-bifurcation scenario of the two-period quasiperiodic response shows that it can undergo a *homoclinic bifurcation*, beyond which only periodic solutions are exhibited by the beam.

Moreover, a distinguishing feature of the investigated interaction is the existence of isolated solution branches, a phenomenon known as *frequency island generation*. Further, *mode localization*—a phenomenon that already occurs within the linear regime—is here manifested in a peculiar fashion as a result of the non-linear coupling. Contrary to the typical normal mode localization referring to confinement of vibrational energy within rather small regions of the disordered systems, in the present case, it is shown that the system may respond, depending on the initial conditions, with a dominant modal contribution from the first mode with most of the vibrational energy confined over three-fifths of the beam span resulting in weakly localized motions. The obtained results are thought to be valuable for the analysis of more general mistuning/asymmetry effects in the design of vibrating beams and structures towards engineering applications.

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Appendix. The linear eigenvalue problem

The c_i and ω in the mode shapes are solutions of the following algebraic eigenvalue problem:

$$\begin{aligned}
&c_1 + c_3 = 0 c_1\omega - c_3\omega + \alpha(c_2\sqrt{\omega} + c_4\sqrt{\omega} + c_5\pi) \\
&\qquad = 0, \\
&c_1 \cos\sqrt{\omega} + c_2 \sin\sqrt{\omega} + c_3 \cosh\sqrt{\omega} \\
&\qquad + c_4 \sinh\sqrt{\omega} = 0, \\
&c_1 \cos\sqrt{\omega} + c_2 \sin\sqrt{\omega} - c_3 \cosh\sqrt{\omega} \\
&\qquad - c_4 \sinh\sqrt{\omega} = 0, \\
&b^2\pi^3\sqrt{\omega}(c_1\Gamma_1 - c_2\Gamma_2 - c_3\Gamma_3 - c_4\Gamma_4) \\
&\qquad + \left[\omega^2 - \pi^4\left(1 + \frac{b^2}{2}\right)\right] = 0,
\end{aligned} \tag{35}$$

where

$$\Gamma_1 = -\frac{\sqrt{\omega}(1 + \cos\sqrt{\omega})}{\pi^2 - \omega^2}, \quad \Gamma_2 = \frac{\sqrt{\omega}\sin\sqrt{\omega}}{\pi^2 - \omega^2}, \tag{36}$$

$$\begin{aligned}
\Gamma_3 &= -\frac{\sqrt{\omega}\left(1 + \cosh\sqrt{\omega}\right)}{\pi^2 + \omega^2}, \\
\Gamma_4 &= -\frac{\sqrt{\omega}\sinh\sqrt{\omega}}{\pi^2 + \omega^2}.
\end{aligned} \tag{37}$$

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